

Natural Computing

LMU Munich
summer term 2025

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*Another digression into
theoretical computer science???*

Who knows about
lazy evaluation?

Who has written an algorithm that
utilizes lazy evaluation?

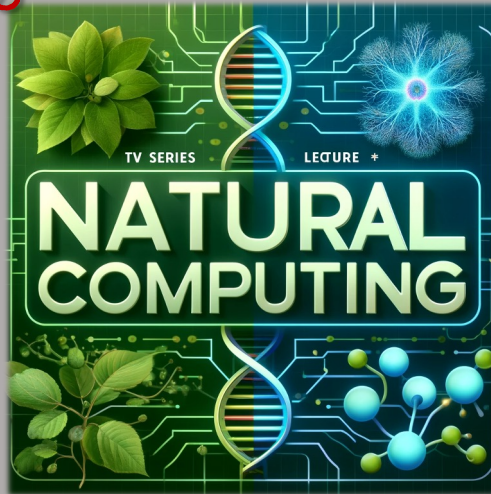
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lazy evaluation

eager/strict evaluation

And now for our main program...

Previously on...



two strange
quantum effects



Superposition



entanglement

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

ASCII letter A

$$q = \sum_{x \in \mathcal{X}} c_x |x\rangle$$

0	0	0	0	0	0	0	0
50%	50%	50%	50%	50%	50%	50%	50%
50%	50%	50%	50%	50%	50%	50%	50%
1	1	1	1	1	1	1	1

all ASCII chars
in superposition



measuring

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

ASCII char E with
probability 1/256

$$q = \sum_{x \in \mathcal{X}} c_x |x\rangle$$

0 90%	0 10%	0 90%	0 90%	0 90%	0 10%	0 90%	0 10%
10% 1	90% 1	10% 1	10% 1	10% 1	90% 1	10% 1	90% 1



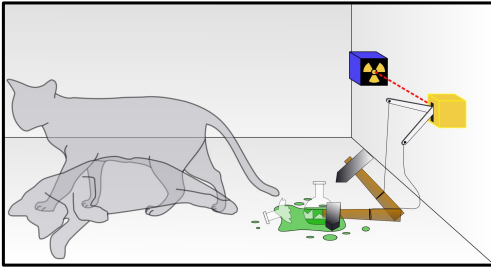
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

all ASCII chars
in superposition
(but probably E)

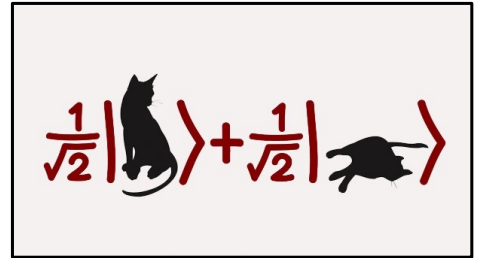
measuring

ASCII char E with
probability 0.43

Schrödinger's Cat



en.wikipedia.org/wiki/Schrödinger%27s_cat

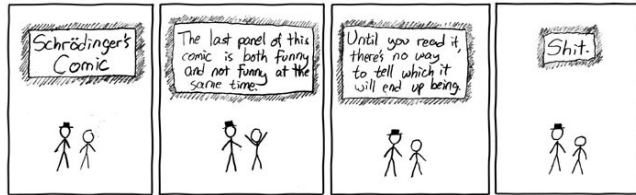


www.motherjones.com/kevin-drum/2018/09/schrodingers-cat-is-alive-one-twelfth-of-the-time/

Schrödinger's Cat



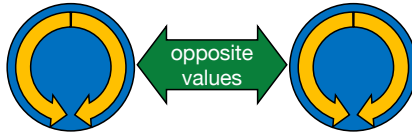
tvtropes.org/pmwiki/pmwiki.php/UsefulNotes/SchrodingersCat



xkcd.com/45



Entanglement



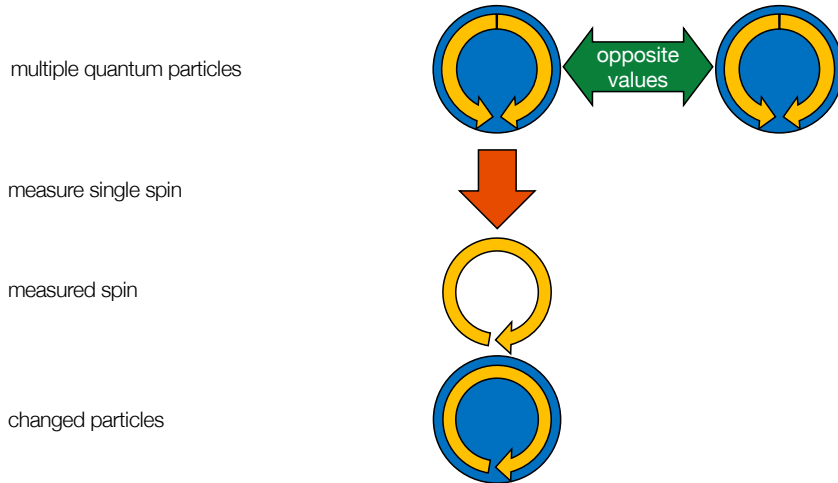
multiple quantum particles

measure single spin

measured spin

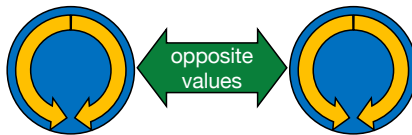
changed particles

Entanglement



Entanglement

multiple quantum particles



measure single spin



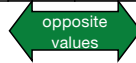
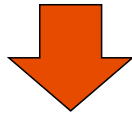
measured spin



changed particles



0	0	0	0	0	0	0	0
1	1	1	1	1	1	50% 1	50% 1



0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

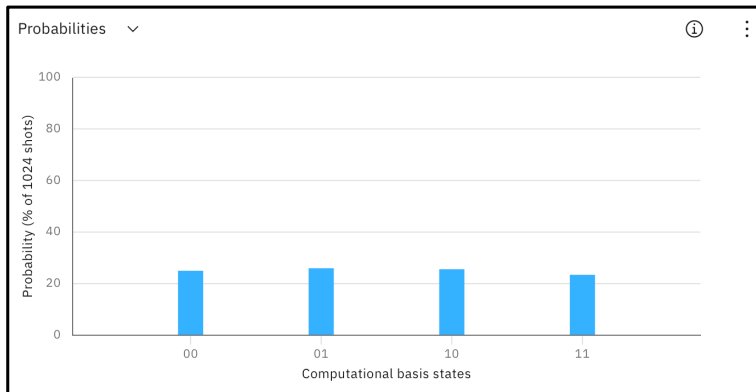
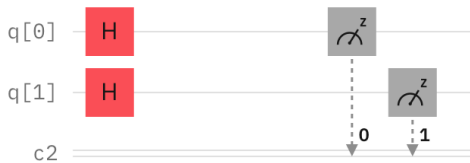
ASCII chars A B at
the same time

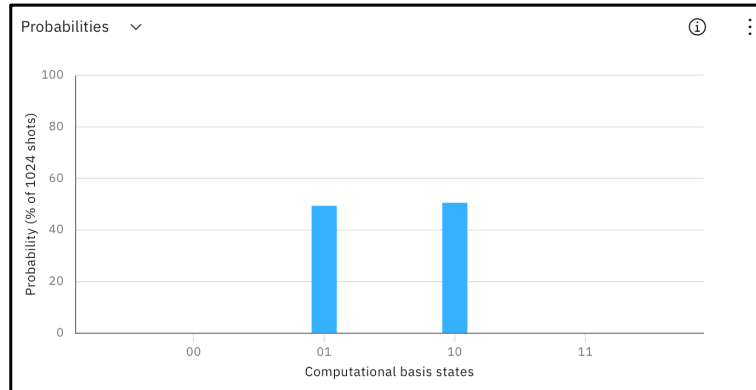
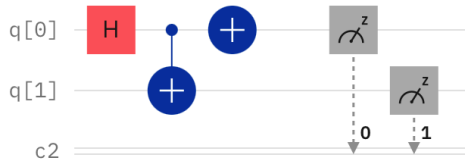
measuring

ASCII char A with
probability 1/2

Let's try

quantum-computing.ibm.com/composer





generated with quantum-computing.ibm.com/composer

Definition 6 (quantum gate). Let $q = \sum_{i=0}^{2^n-1} c_i |i\rangle$ be an n -qubit register. We often write

$$q = \begin{bmatrix} c_{0\dots 0} \\ \vdots \\ c_{1\dots 1} \end{bmatrix}$$

as a shorter notation when the basis $|0\dots 0\rangle, \dots, |1\dots 1\rangle$ is fixed.

An n -ary quantum gate is an operator that works on an n -qubit register q , i.e., (possibly) alters its superposition amplitudes, and thus produces a new n -qubit register q' . An n -ary quantum gate can be given via a matrix $U \in \mathbb{C}^{n \times n}$ so that

$$q' = Uq.$$

Quantum gates commonly implemented on quantum computers are unitary, which — among other things — implies they are reversible.

A little excursion:

What happens to computers
when they compute a lot?

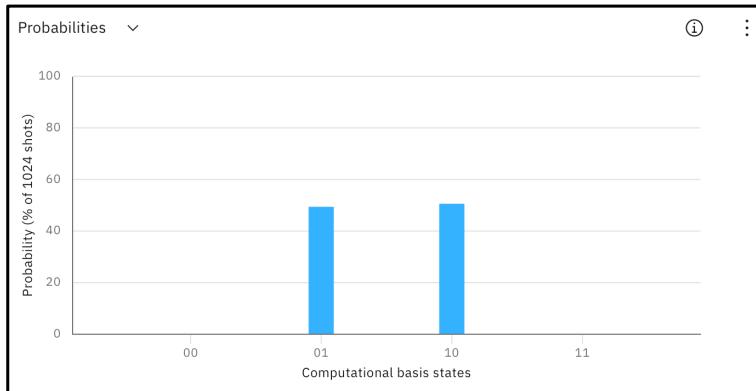
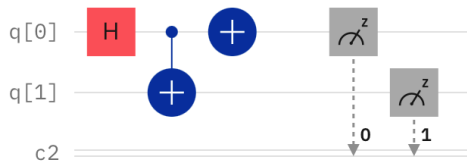
A little excursion:

Why do computers get hot
when they compute?

A little excursion:

Why are most quantum operators
reversible?

Operator	Gate(s)		Matrix
Pauli-X (X)			$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)			$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)			$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)			$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)			$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)			$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)			$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$



generated with quantum-computing.ibm.com/composer

Quantum Advantage

One more time!

Theorem 1 (Church-Turing thesis). Any computable function can be computed by a Turing machine.

Theorem 2 (extended Church-Turing thesis). Any computable function can be computed by a Turing machine *with at most polynomial overhead*.

Quantum Advantage?

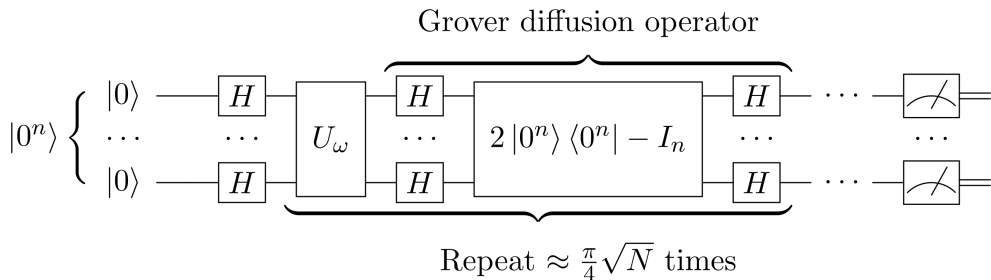
Grover's Search

Problem: Search in unsorted data structure with N elements

Classical Complexity: $\mathcal{O}(N)$

Quantum Complexity: $\mathcal{O}(\sqrt{N})$

en.wikipedia.org/wiki/Grover%27s_algorithm



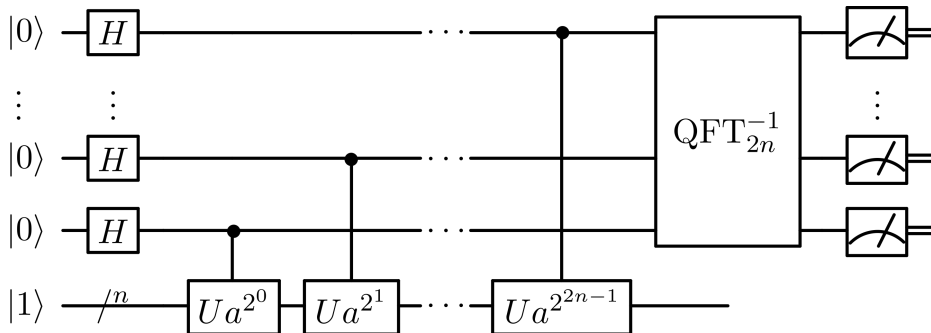
Shor's Algorithm

Problem: Prime factorization of integer N

Classical Complexity: best-known exponential in $\log N$

Quantum Complexity: polynomial in $\log N$

en.wikipedia.org/wiki/Shor%27s_algorithm



HHL Algorithm

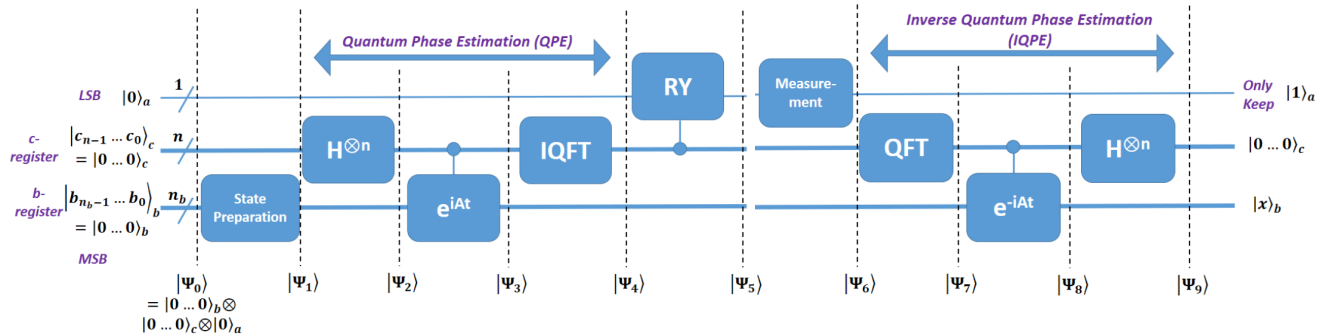
named after
Harrow, Hassidim, Lloyd

Problem: Solving linear systems of N equations with condition number κ

Classical Complexity: $\mathcal{O}(\sqrt{\kappa}N)$ for a scalar value computed from solution

Quantum Complexity: $\mathcal{O}(\kappa^2 \log N)$ for a scalar value computed from solution

adapted from arxiv.org/pdf/2108.09004.pdf



Quantum Annealing

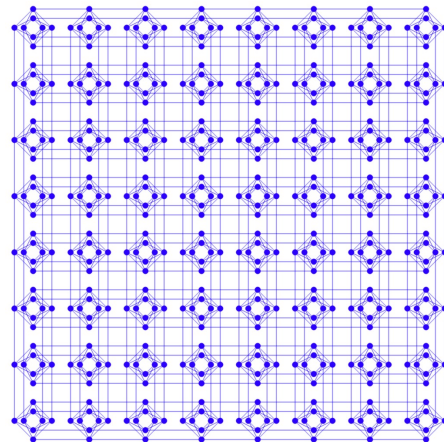
introduced by
Kadowaki, Nishimori

Problem: Solving (quadratic unconstrained binary) optimization problems

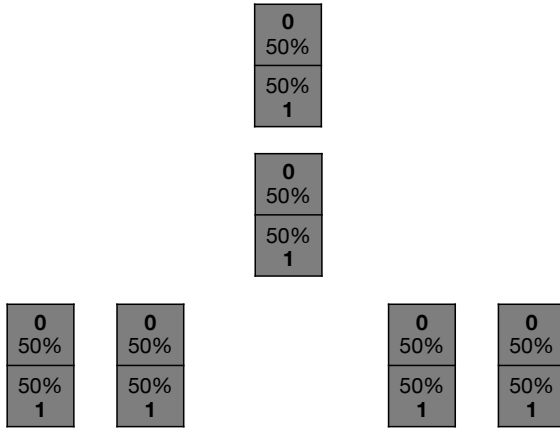
Classical Complexity: best-known exponential for exact solution

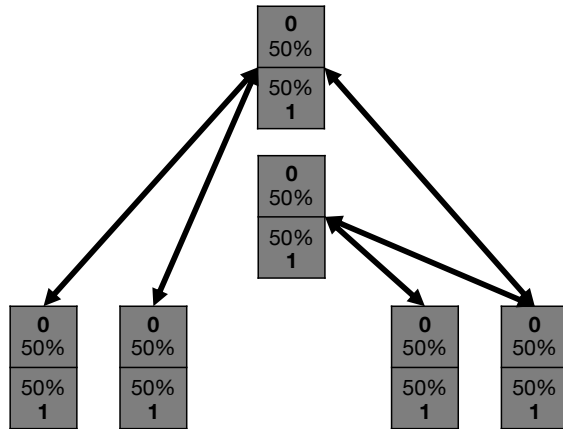
Quantum Complexity: probably exponential as well?

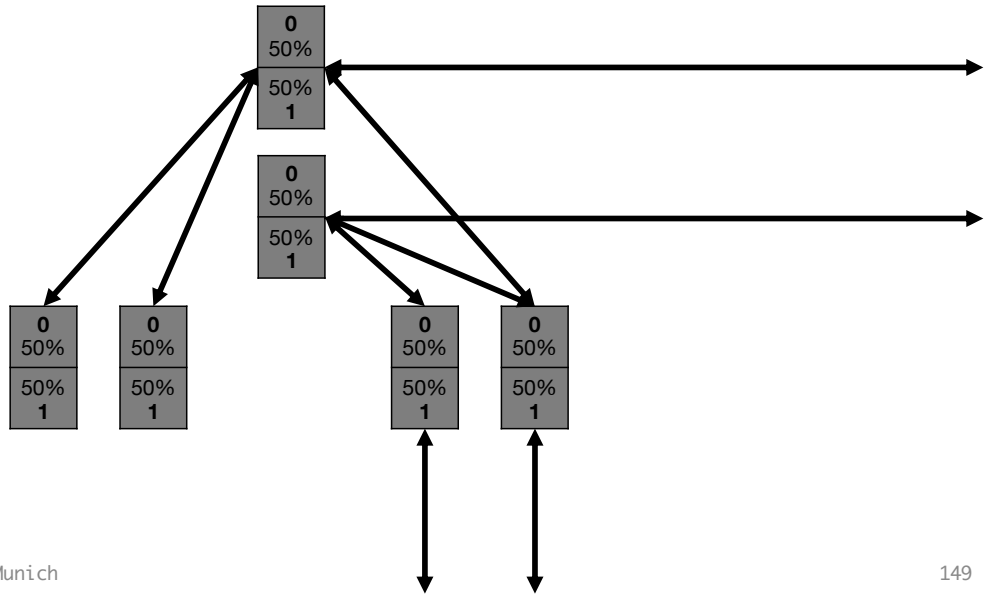
arxiv.org/pdf/1406.2741.pdf

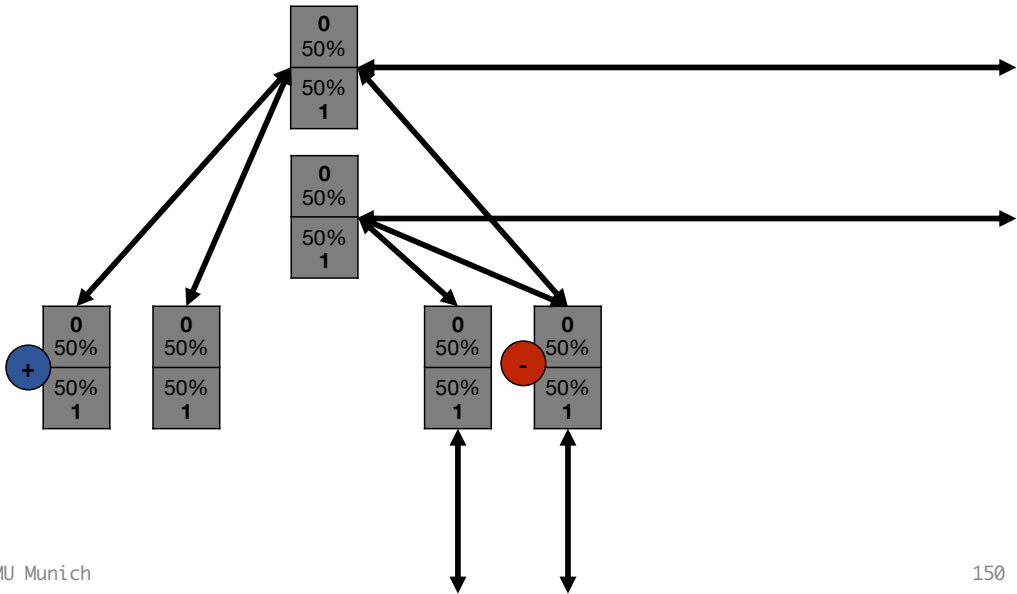


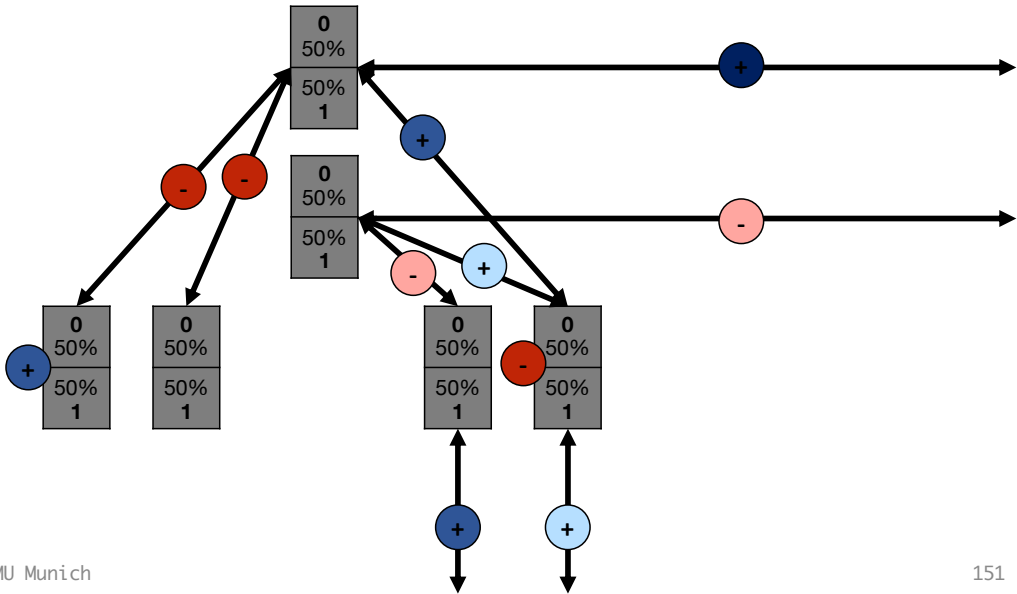
So what is
quantum annealing?





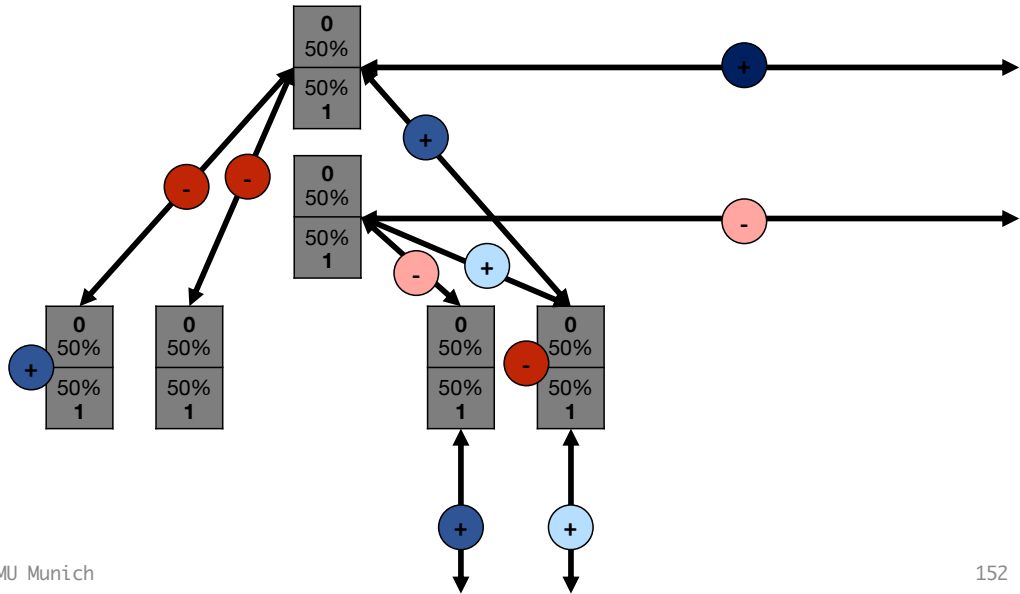




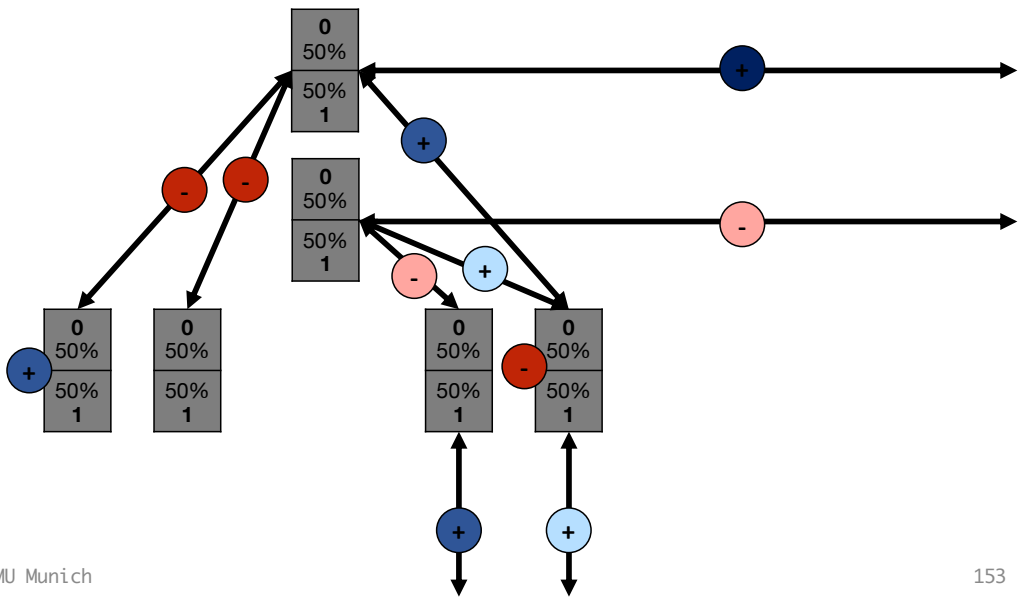


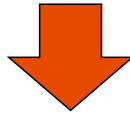
adiabatic quantum computing

add constraints
infinitely slowly
while adding
no energy

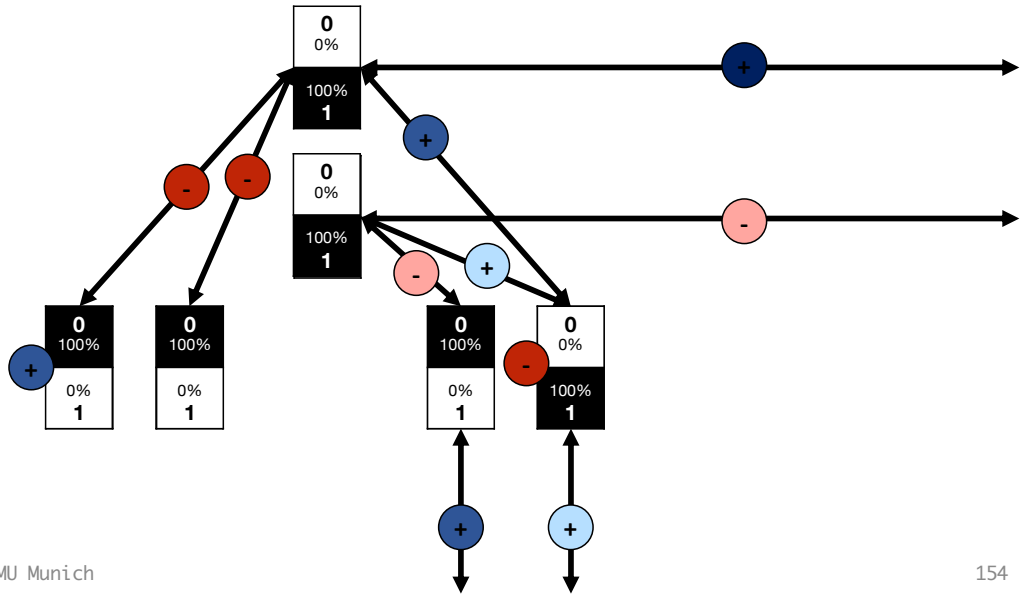


practical quantum annealing





measuring



Definition 7 (Ising model). Let $G = (V, E)$ be a graph with vertices $V = \{v_0, \dots, v_{n-1}\}$ for $n \in \mathbb{N}$ and undirected edges $E \subseteq V \times V$. A spin configuration $x \in \mathcal{X} = (V \rightarrow \{-1, +1\})$ is a mapping of vertices to values $\{-1, +1\}$. W.l.o.g. we assume that x has the form $x(v_i) = \sigma_i$ for any $i \in \{0, \dots, n-1\}$ and a spin vector $\vec{\sigma} \in \{-1, +1\}^n$. The energy of an Ising model is then given via

$$H(\vec{\sigma}) = - \sum_{\substack{i \in \{0, \dots, n-1\} \\ j \in \{0, \dots, n-1\}}} J_{ij} \sigma_i \sigma_j - \sum_{i \in \{0, \dots, n-1\}} h_i \sigma_i$$

where $J \in \mathbb{R}^{n \times n}$ is called the interaction matrix and $\vec{h} \in \mathbb{R}^n$ is called the vector of field strengths. We also assume that for any $i, j \in \{0, \dots, n-1\}$ it holds that $i \geq j \implies J_{ij} = 0$, i.e., J is a strictly upper triangle matrix, and that $(v_i, v_j) \notin E \implies J_{ij} = 0$, i.e., J_{ij} can only have non-zero values where an edge between i and j exists in the substrate graph G .

For an Ising model given by (G, J, h) , there are processes (both artificial and natural) that produce a spin configuration $\vec{\sigma}$ so that $H(\vec{\sigma})$ is minimized.

Definition 8 (QUBO). Let $Q \in \mathbb{R}^{n \times n}$ be an arbitrary matrix called QUBO matrix. The process of finding a vector $\vec{x} \in \mathbb{B}^n$ so to

$$\text{minimize } H(\vec{x}) = x^\top Q x = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} Q_{ij} x_i x_j$$

is called *quadratic unconstrained binary optimization (QUBO)*. W.l.o.g. we can assume that Q is a (non-strictly) upper triangular matrix.

Ising–QUBO Equivalence